YASH RAJ PANDEY

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Summary

AI systems engineer building local-first LLM infrastructure, RAG and agent platforms, and production evaluation systems. Promoted from Software Engineer to AI Agents Architect at UF in 13 months; shipped systems serving 30+ researchers and 5M+ records, with 45+ merged PRs across 30+ open-source projects.

Experience

University of Florida — **Gainesville, FL**

AI Agents Architect — *Apr 2026 – Present*

- Architect and deploy a production AI agent platform combining RAG over 10,000+ technical documents with tool-calling workflows and natural-language access to research data.
- Built a model-selection and release-gating system spanning 14 LLMs, 26 task-specific evaluations, and a 24-case regression suite; identified an 88% multi-tool orchestration leader and caught hallucination failures before deployment.
- Lead the migration to self-hosted inference on institutional GPUs for a ∼745B-parameter open-weight MoE (FP8, 1M-token context), projected to reduce 5-year TCO ∼60% and per-query cost to ∼$0.01 while keeping sensitive data on-premises.
- Migrated 50,000+ records to DuckDB and audited 1.4M+ cells with zero mismatches, surfacing 2,000+ ID conflicts before migration lock-in.

Lead Software Engineer — *Oct 2025 – Apr 2026*

- Led Blue Omics, a Django, React, and PostgreSQL platform that became the system of record for 30+ researchers across 5 labs; scaled it to 5.38M+ records and 40% growth in daily active users.
- Reduced core lookup latency to 23–29 ms, analytics to ∼222 ms over ∼10K records, and BLASTN search to ∼602 ms on a 518 MB index.
- Shipped 7 ingestion and validation pipelines, cut end-to-end R Markdown reporting from 2–3 hours to 15–20 minutes, deployed on GCP, and grew the team from one to three.

Software Engineer — *Mar 2025 – Oct 2025*

- Designed the original PostgreSQL schemas, Python ingestion, and validation logic; cut frontend load time from 8s to 3s and established the foundation later scaled to 5M+ records.

Selected Projects

Looma | *Local-first memory for coding agents* — *github.com/devYRPauli/looma*

- Built a zero-dependency Python CLI that converts Claude Code, Codex, and Cursor transcripts into structured work items and token-budgeted context packs using SQLite and FTS5; 134 passing tests.
- Reached F1 0.86 (0.94 precision) with heuristic extraction and 0.95 with an optional local LLM; sped ingestion ∼27x and improved retrieval MRR 49%.

mddocs | *Git-native collaborative Markdown with an agent API* — *github.com/devYRPauli/mddocs*

- Published an MIT-licensed collaborative Markdown editor with Yjs multiplayer and a token-gated agent API; shipped 8 npm releases and cut package size 87% with esbuild.

Independent Evaluations | *TurboQuant KV-cache compression and Google TabFM* — *yashrajpandey.com/writing*

- Fixed five root causes in TurboQuant implementations across MLX and llama.cpp forks, restoring needle retrieval from 0% to 100% at 16K tokens on a 16 GB M1 Pro with 3.5x less KV-cache memory.
- Reproduced Google TabFM across 3 machines and 13 TabArena datasets; beat tuned XGBoost on all 10 fold-matched datasets and contributed a merged multi-GPU crash fix to google-research/tabfm.

Technical Stack

AI Systems:	LLM Inference and Evals, RAG, Agent Runtimes, Tool-Calling, vLLM, Ollama, llama.cpp, MLX, Quantization, Vector Search, Embeddings, Reranking, MCP
Engineering:	Python, TypeScript, JavaScript, SQL, C/C++, Django, FastAPI, React, Next.js, PostgreSQL, DuckDB, Redis, SQLite
Infrastructure:	Docker, Kubernetes, Terraform, GCP, AWS, Linux, GPU / Self-Hosted Inference, Git, GitHub Actions

Education

University of Florida — **Aug 2023 – Dec 2024**

M.S. Computer & Information Science & Engineering (GPA 3.8) — *Gainesville, FL*

Jaypee University of Engineering and Technology — **Jul 2019 – May 2023**

B.Tech. in Computer Science and Engineering — *Guna, India*